

# Does your design need analog intelligence from an 8-bit microcontroller?

PIC16F753 enables intelligent sensing plus efficient power conversion and management

## Low-Cost PIC® MCU with Intelligent Analog Small Form Factor



Boost system performance and efficiency, while reducing system costs, with the enhanced analog capabilities of the PIC16F753 microcontroller. With its integrated Op Amp, slope compensation and range of core-independent peripherals, the PIC16F753 enables efficient power conversion and provides a versatile platform for adding intelligent sensing and power management to embedded designs.

Microchip's PIC16F753 8-bit microcontroller offers all the key features of the PIC12F752 and adds an Op Amp with 3 MHz of gain bandwidth product, in addition to slope compensation for switch-mode power supplies.

Non-overlapping, complementary waveforms, for comparator and PWM inputs, are provided by the on-chip complementary output generator, with dead-band, phase and blanking control, as well as auto shutdown and reset. The on-chip 8-channel, 10-bit ADC can be used to add sensing capabilities, including capacitive touch user interfaces, and two 50 mA outputs support direct driving of FETs. Also on-board are high-performance comparators, a 9-bit DAC, and a capture compare PWM module.

For high-voltage cost-sensitive designs, the PIC16HV753 features a shunt regulator for operation from 2V to a user-defined maximum voltage level, with less than 2 mA operation current.

### GET STARTED TODAY!

1. Free reference design for a high-power LED flashlight
2. Choice of low-cost development tools:
  - PICDEM™ Lab Development Kit
  - PICKIT™ 3
  - PICKIT Low-Pin Count Demo Board

[www.microchip.com/intelligentanalog](http://www.microchip.com/intelligentanalog)



PICDEM™ Lab Development Kit  
(DM163045)

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